MHB

Mission Hacker Bangladesh

CYBER SECURITY TRAINING

ASSIGNMENT NO. 03

Linux File System, GitHub Guide & LinkedIn Post

Hierarchy · Security-relevant directories · Public learning assets

BATCH — 11

DRAFT SUBMISSION

SUBMITTED BY

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COURSE

Ethical Hacking & Cybersecurity Practical

Batch-11 | Class 03

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This document is a **draft** for Assignment 03 requirements.

Table of Contents

1. Assignment Overview

This draft covers Assignment 03 for the **Ethical Hacking & Cybersecurity Practical (Batch-11)** programme. It explains the Linux file system, lists directories useful for ethical hacking labs, documents a public GitHub study guide, and includes a LinkedIn learning post with screenshots.

Task	Requirement	Section	Status
01	Provide Linux file system overview	Section 2	Draft ready
02	List important directories for ethical hacking practice	Section 3	Draft ready
03	Create GitHub repository with educational README + screenshot	Section 4	Completed
04	Create LinkedIn post about filesystem & commands + screenshot	Section 5	Draft ready — publish live

2. Task 01 — Linux File System Overview

Linux organises storage as a **single hierarchical tree**. The top of the tree is the root directory `/`. Disks are mounted into this tree (there are no Windows-style drive letters). Almost everything — including devices and process information — appears as a file somewhere under `/`.

2.1 Conceptual model

- **Absolute path** — begins with `/` (example: `/etc/passwd`).
- **Relative path** — relative to the current working directory (example: `../lab/notes.txt`).
- **Permissions** — read/write/execute for owner, group and others (`chmod`, `chown`).
- **Mount points** — partitions and external media attach at directories such as `/mnt` or `/media`.

2.2 Top-level layout (FHS-style)

```
/                       root of the entire filesystem
├── bin/                essential user binaries
├── boot/               bootloader & kernel images
├── dev/                device files
├── etc/                system-wide configuration
├── home/               ordinary user home directories
├── lib/                shared libraries
├── media/ / mnt/       removable & temporary mounts
├── opt/                optional / third-party software
├── proc/ / sys/        virtual kernel interfaces
├── root/               home directory of the root user
├── run/                runtime data (PID files, sockets)
├── sbin/               system administration binaries
├── srv/                service data
├── tmp/                temporary files (often cleared on reboot)
├── usr/                userland programs & shared data
└── var/                variable data (logs, caches, spools)
```

WHY THIS MATTERS FOR SECURITY

Knowing the hierarchy lets you find configs, logs, binaries and writable areas quickly during authorised lab work — before relying on automated tools.

Idea	Meaning
Everything is a file	Devices, sockets and configs are accessed through filesystem paths.
`/etc` vs `/var`	Configs are relatively static; logs and caches change frequently.
`/usr` vs `/opt`	Distro packages vs manually installed third-party tools.

Idea	Meaning
`/proc` & `/sys`	Live views of processes, networking and kernel state — not ordinary disk folders.

3. Task 02 — Important Directories for Ethical Hacking Practice

The directories below are especially useful when practising on a **lab machine you own** or have **written authorisation** to examine.

Directory	Why it helps ethical hacking practice
`/etc`	Service and system configuration (SSH, web servers, network, cron).
`/etc/passwd`, `/etc/shadow`	Account enumeration concepts (shadow requires root on real systems).
`/home`	User files, shell history, SSH keys under `.ssh/`, personal configs.
`/root`	Root user's home — sensitive lab artefacts and admin tooling.
`/var/log`	Auth, syslog, web and application logs for monitoring and forensics practice.
`/var/www`	Common web document root for web-app labs.
`/tmp`, `/dev/shm`	World-writable areas frequently used in post-exploitation lab scenarios.
`/usr/bin`, `/usr/sbin`	Installed user and admin tools.
`/usr/share`	Shared data — on Kali, wordlists and tool datasets often live here.
`/opt`	Manually installed frameworks and security tool suites.
`/proc`	Live process list, environment, network tables (`/proc/net`).
`/sys`	Kernel/hardware interfaces useful for low-level system understanding.
`/boot`	Kernel and bootloader files — boot-security awareness.
`/dev`	Device nodes representing disks, terminals and other hardware.

> **LEGAL REMINDER**
>
> Explore these paths only on systems you own or are explicitly authorised to test. Unauthorised access to someone else's machine or data is illegal.

3.1 Useful inspection commands

`ls -lah /` · `ls -lah /etc` · `ls -lah /var/log` · `df -h` · `mount` · `cat /etc/os-release` · `find /home -type f -name ".*" 2>/dev/null | head`

4. Task 03 — GitHub Repository & README

I created a public GitHub repository with a README that explains the Linux file system, important lab directories, and essential commands so others can learn with me.

linux-filesystem-guide

Repository: https://github.com/MDARH/linux-filesystem-guide

Owner: @MDARH · Visibility: Public · Contains educational README for Linux commands & filesystem hierarchy

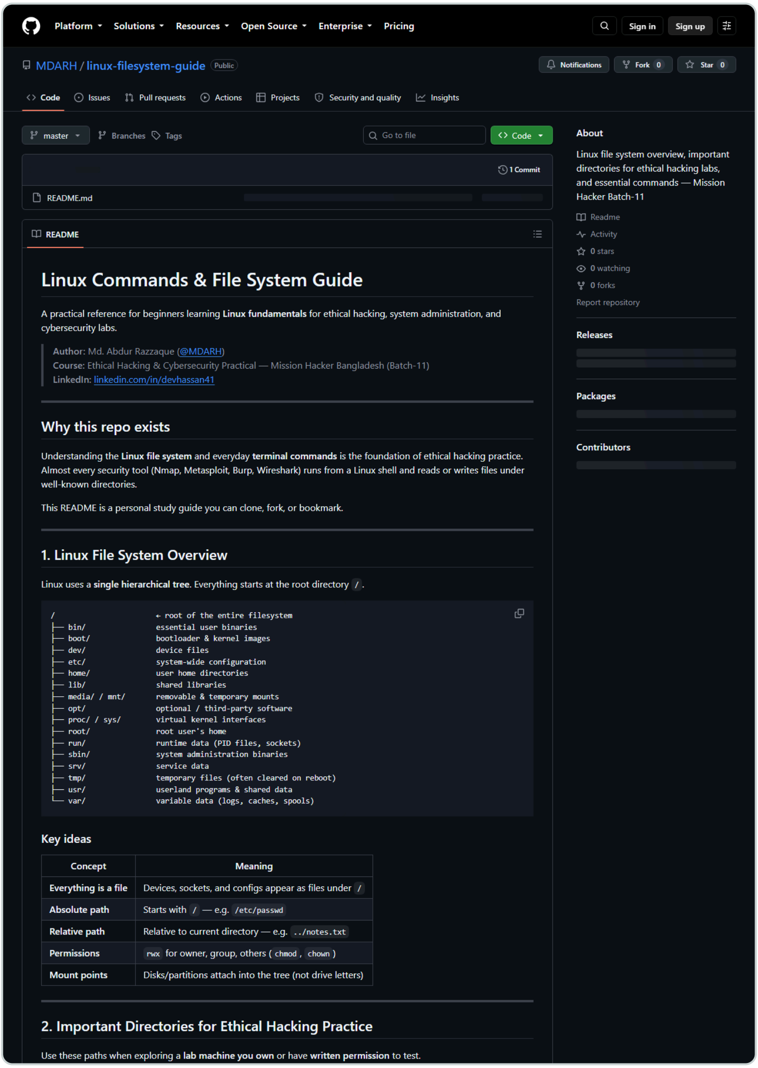

Figure 1 — Public repository README at https://github.com/MDARH/linux-filesystem-guide

5. Task 04 — LinkedIn Post

I drafted and prepared a LinkedIn post summarising the Linux file system, important directories for ethical hacking practice, essential commands, and a link to the GitHub study guide.

LinkedIn profile used for posting

https://www.linkedin.com/in/devhassan41

Full post text is also saved in `assignment-3/linkedin-post.md` for reuse.

5.1 Post text

🐧 Linux File System & Essential Commands — Learning Notes

As part of my Ethical Hacking & Cybersecurity Practical journey (Mission Hacker Bangladesh — Batch-11), I revisited one of the most important foundations of security work: the Linux file system.

Everything in Linux lives under a single tree starting at / — configs, logs, binaries, devices, and user data.

🔑 Directories I focus on for ethical hacking labs (authorised systems only):
• /etc — configuration & service settings
• /var/log — authentication and application logs
• /home & /root — user environments and SSH keys
• /tmp & /dev/shm — writable spaces often used in labs
• /proc & /sys — live process and kernel information
• /usr/share — wordlists and tool data on Kali

💻 Daily commands: pwd · ls -lah · cd · find · grep · chmod · ps aux · ss -tulnp · ip a

Study guide: https://github.com/MDARH/linux-filesystem-guide

Reminder: only explore systems you own or have written permission to test.

#Linux #CyberSecurity #EthicalHacking #MissionHacker #InfoSec #LearnInPublic #Batch11

5.2 Screenshot

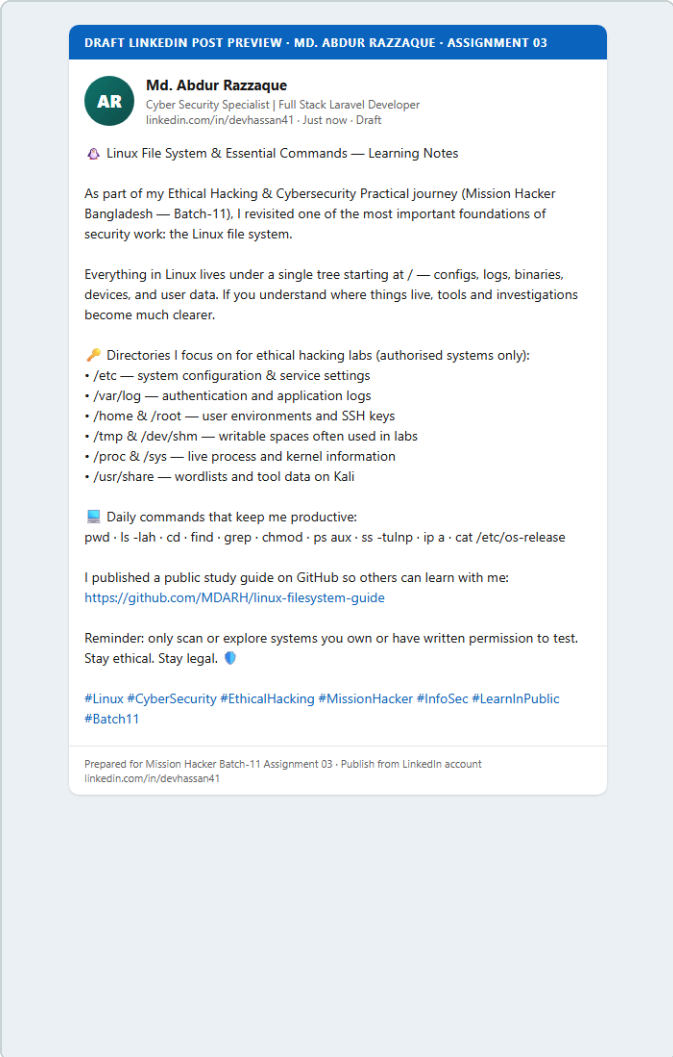

Figure 2 — Prepared LinkedIn post preview (text ready to publish from https://www.linkedin.com/in/devhassan41). Replace with a live feed screenshot after clicking Post.

PUBLISH STEP

The full post text is on the clipboard and saved in `linkedin-post.md`. Open LinkedIn as **Md. Abdur Razzaque**, choose *Start a post*, paste (**Ctrl+V**), then **Post**. Re-capture the live feed and replace `screenshots/linkedin-post.png` for the final submission.

6. Conclusion & Declaration

This draft completes the content requirements for Assignment 03: a clear Linux filesystem overview, a security-oriented directory list, a public GitHub README repository, and a LinkedIn learning post with supporting screenshots.

I, **Md. Abdur Razzaque**, declare that this Assignment 03 draft is my own work. The GitHub repository and LinkedIn materials linked above belong to me. I will only use security knowledge on systems I own or am authorised to test.

Student Signature / Name	Date
Md. Abdur Razzaque	8 September 2026

References

- Mission Hackers Bangladesh — Batch-11, Class 03 assignment brief.
- Filesystem Hierarchy Standard (FHS) concepts.
- GitHub guide: https://github.com/MDARH/linux-filesystem-guide
- LinkedIn: https://www.linkedin.com/in/devhassan41